

Normal distribution

$$\mathcal{N}_{\mu,\sigma}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

CDF of normal distribution (term 1/2 because erf integration starts from 0)

$$\mathcal{C}_{\mu,\sigma}(x) = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sqrt{2}\sigma}\right) \right]$$

Finding n_σ from Δx Extract probability for minimal one-sided deviation $\Delta x = x - \mu$ and standard deviation σ from mean of normal distribution

$$p = \operatorname{erf}\left(\frac{\Delta x}{\sqrt{2}\sigma}\right)$$

Rewrite a probability p (close to 1 = large n_σ) into number of sigmas n_σ

$$n_\sigma = \sqrt{2} \operatorname{erfinv}(p)$$

Finding Δx from n_σ Just invert equations from above

$$p = \operatorname{erf}\left(\frac{n_\sigma}{\sqrt{2}}\right), \quad \Delta x = \sqrt{2}\sigma \operatorname{erfinv}(p)$$